

# Lecture 7: Normal Distribution, Joint PDFs

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## 1 Normal Distribution

### 1.1 Why the normal distribution?

Thus far, we've seen two continuous distributions – the **Uniform** (equally likely along an interval) and the **Exponential** (waiting times, memorylessness).

The **Normal** (or **Gaussian**) distribution is a little different than these – it arises naturally as the *universal limit* of averaging many small, independent effects — measurement error, heights, exam scores, noise in a communication channel. We'll make this precise later in the course (the **Central Limit Theorem**).

It is (one of) the most important distribution(s) in probability and statistics.

### 1.2 Normal random variables: definition

**i** Definition: Normal random variable

A continuous random variable  $X$  is **normal** (or **Gaussian**) with parameters  $\mu \in \mathbb{R}$  and  $\sigma^2 > 0$ , written  $X \sim N(\mu, \sigma^2)$ , if it has PDF

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

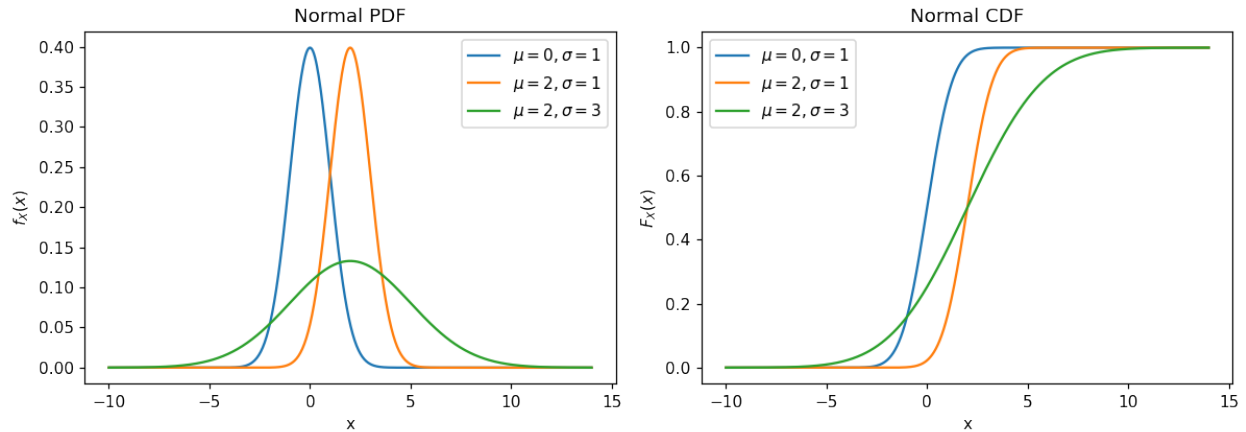
- The density is a symmetric “bell curve” centered at  $\mu$ .
- $\sigma$  controls the spread: small  $\sigma$  gives a tall, narrow bump; large  $\sigma$  gives a short, wide one.
- It’s not obvious from the formula alone that this integrates to 1 — that relies on the classical **Gaussian integral**  $\int_{-\infty}^{\infty} e^{-u^2/2} du = \sqrt{2\pi}$ , which we’ll take as known and verify numerically below.

---

```
# Visualizing the PDF and CDF for a few choices of (mu, sigma)
xs = np.linspace(-10, 14, 500)
params = [(0, 1), (2, 1), (2, 3)]

fig, axes = plt.subplots(1, 2, figsize=(11, 4))
for mu, sigma in params:
    axes[0].plot(xs, stats.norm.pdf(xs, mu, sigma), label=f"$\mu={mu}, \sigma={sigma}$")
    axes[1].plot(xs, stats.norm.cdf(xs, mu, sigma), label=f"$\mu={mu}, \sigma={sigma}$")

axes[0].set_title("Normal PDF"); axes[0].set_xlabel("x"); axes[0].set_ylabel(r"$f_X(x)$"); axes[0].legend()
axes[1].set_title("Normal CDF"); axes[1].set_xlabel("x"); axes[1].set_ylabel(r"$F_X(x)$"); axes[1].legend()
plt.tight_layout()
plt.show()
```



### 1.2.1 Mean and variance of normal RV

#### 💡 Fact

If  $X \sim N(\mu, \sigma^2)$ , then

$$E[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

$E[X] = \mu$ : substitute  $u = x - \mu$ . The density becomes symmetric in  $u$  about 0, i.e.  $f_X(\mu + u) = f_X(\mu - u)$ , and

$$E[X] - \mu = E[X - \mu] = \int_{-\infty}^{\infty} u \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-u^2/(2\sigma^2)} du = 0,$$

because the integrand is an **odd function** of  $u$ .

$\text{Var}(X) = \sigma^2$ : this takes an integration-by-parts argument on  $\int u^2 e^{-u^2/(2\sigma^2)} du$ .

---

### 1.2.2 Normality is preserved by linear transformations

#### **i** Theorem

If  $X \sim N(\mu, \sigma^2)$  and  $Y = aX + b$  for constants  $a \neq 0, b$ , then

$$Y \sim N(a\mu + b, a^2\sigma^2).$$

Proof

Recall if  $g$  is invertible and differentiable with inverse  $h = g^{-1}$ , then  $f_Y(y) = f_X(h(y)) |h'(y)|$ . Here  $g(x) = ax + b$ , so  $h(y) = (y - b)/a$  and  $h'(y) = 1/a$ . Then

$$f_Y(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}\right) \cdot \frac{1}{|a|} = \frac{1}{|a|\sigma\sqrt{2\pi}} \exp\left(-\frac{(y - (a\mu + b))^2}{2a^2\sigma^2}\right),$$

which is exactly the  $N(a\mu + b, a^2\sigma^2)$  density. ■

A linear function of a normal random variable is *again normal* — this is a special structural property that most distributions don't have (e.g., a linear function of a Uniform or Exponential RV is generally *not* Uniform or Exponential).

### 1.3 The standard normal random variable

#### **i** Definition: Standard normal

$Z \sim N(0, 1)$  is the **standard normal** random variable. Its PDF and CDF are denoted by lower- and uppercase phis:

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad \Phi(z) = \int_{-\infty}^z \varphi(t) dt.$$

$\Phi$  has no closed form in terms of elementary functions — historically this is why every probability textbook has a **standard normal table**:

| $z$       | 0.00   | 0.50   | 1.00   | 1.50   | 1.645  | 2.00   |
|-----------|--------|--------|--------|--------|--------|--------|
| $\Phi(z)$ | 0.5000 | 0.6915 | 0.8413 | 0.9332 | 0.9500 | 0.9772 |

You can use `scipy.stats.norm` to play the role of the table, but the *method* — standardize, then look up — is exactly what you'd do with a printed table.

#### 1.3.1 Standardizing a normal random variable

#### **i** Corollary (of the linear-transformation theorem)

If  $X \sim N(\mu, \sigma^2)$ , then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

This is immediate from the theorem two slides ago with  $a = 1/\sigma$ ,  $b = -\mu/\sigma$ :  $a\mu + b = 0$  and  $a^2\sigma^2 = 1$ .

This lets us answer probability questions about *any* normal  $X \sim N(\mu, \sigma^2)$  using only the single function  $\Phi$ !

$$P(X \leq x) = P\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$


---

### 1.3.2 Example: snowfall

Annual snowfall  $X$  (in inches) at a mountain resort is modeled as  $X \sim N(60, 20^2)$ .

(a) What's the probability of more than 90 inches of snow this year?

$$P(X > 90) = 1 - \Phi\left(\frac{90 - 60}{20}\right) = 1 - \Phi(1.5).$$

(b) How much snow is exceeded in only 5% of years — i.e. find  $x^*$  with  $P(X > x^*) = 0.05$ ?

$$1 - \Phi\left(\frac{x^* - 60}{20}\right) = 0.05 \iff \Phi\left(\frac{x^* - 60}{20}\right) = 0.95 \iff \frac{x^* - 60}{20} = 1.645$$

using the table value  $\Phi(1.645) \approx 0.95$  — so  $x^* \approx 60 + 20(1.645) = 92.9$  inches.

```
mu, sigma = 60.0, 20.0

# (a)
p_over_90 = 1 - stats.norm.cdf((90 - mu) / sigma)
print(f"(a) P(X > 90) = {p_over_90:.4f}    [table method: 1 - Phi(1.5) = {1 - 0.9332:.4f}]")

# (b) 95th percentile
z95 = stats.norm.ppf(0.95)
x_star = mu + sigma * z95
print(f"(b) z with Phi(z)=0.95: {z95:.3f}")
print(f"    x* (95th percentile) = {x_star:.2f} inches")
```

(a)  $P(X > 90) = 0.0668$  [table method:  $1 - \Phi(1.5) = 0.0668$ ]

(b)  $z$  with  $\Phi(z)=0.95$ : 1.645

$x^*$  (95th percentile) = 92.90 inches

---

### 1.3.3 Example: signal detection

A transmitter sends a bit  $s \in \{-1, +1\}$  (equally likely) over a noisy channel. The receiver observes

$$Y = s + N, \quad N \sim N(0, \sigma^2) \text{ independent of } s,$$

and decides “+1 was sent” if  $Y > 0$ , and “−1 was sent” otherwise.

#### Question

What is the probability the receiver makes an error?

By symmetry it suffices to compute the error probability given  $s = +1$  was sent (the case  $s = -1$  is identical by symmetry of the noise):

$$P(\text{error} \mid s = 1) = P(Y < 0 \mid s = 1) = P(1 + N < 0) = P(N < -1) = \Phi\left(\frac{-1}{\sigma}\right).$$

So  $P(\text{error}) = \Phi(-1/\sigma)$  — smaller noise  $\sigma$  (relative to the signal strength 1) means a more reliable channel, exactly as intuition demands.

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```

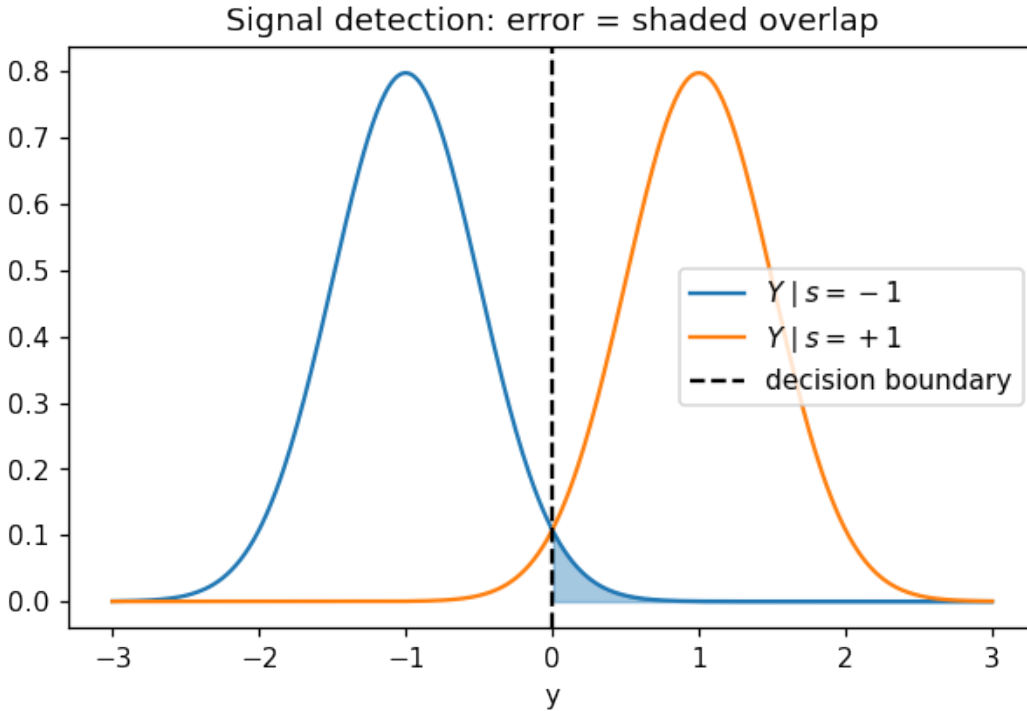
sigma = 0.5
p_error_formula = stats.norm.cdf(-1 / sigma)
print(f"P(error) = Phi(-1/sigma) = {p_error_formula:.4f}    (sigma = {sigma})")

# Verify by simulation
n = 2_000_000
s = rng.choice([-1, 1], size=n)
N = rng.normal(0, sigma, size=n)
Y = s + N
decision = np.where(Y > 0, 1, -1)
p_error_sim = np.mean(decision != s)
print(f"P(error), simulated = {p_error_sim:.4f}")

# Visualize the two conditional densities and the decision boundary
y_grid = np.linspace(-3, 3, 400)
plt.figure(figsize=(6.5, 4))
plt.plot(y_grid, stats.norm.pdf(y_grid, -1, sigma), label=r"$Y \mid s=-1$")
plt.plot(y_grid, stats.norm.pdf(y_grid, 1, sigma), label=r"$Y \mid s=+1$")
plt.axvline(0, color="k", linestyle="--", label="decision boundary")
plt.fill_between(y_grid[y_grid > 0], 0, stats.norm.pdf(y_grid[y_grid > 0], -1, sigma),
                 alpha=0.4, color="C0")
plt.legend(); plt.xlabel("y"); plt.title("Signal detection: error = shaded overlap")
plt.show()

P(error) = Phi(-1/sigma) = 0.0228    (sigma = 0.5)
P(error), simulated = 0.0228

```



## 2 Joint PDFs

### 2.1 Joint PDF of continuous random variables

#### **i** Definition: Joint PDF

Random variables  $X, Y$  are **jointly continuous** with **joint PDF**  $f_{X,Y}(x, y)$  if  $f_{X,Y}(x, y) \geq 0$  everywhere,

$$\iint_{\mathbb{R}^2} f_{X,Y}(x, y) dx dy = 1,$$

and for every (nice) region  $B \subseteq \mathbb{R}^2$ ,

$$P((X, Y) \in B) = \iint_B f_{X,Y}(x, y) dx dy.$$

#### **i** Definition: Marginal PDF

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx.$$

These are the continuous analogs of the discrete cases – sums become integrals!

#### 2.1.1 Example: a 2D uniform density

Let  $(X, Y)$  be **uniform on the unit disk**  $D = \{(x, y) : x^2 + y^2 \leq 1\}$ , i.e.

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{\pi} & (x,y) \in D \\ 0 & \text{otherwise,} \end{cases}$$

since the disk has area  $\pi$ . What is the marginal  $f_X(x)$ ?

$$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} dy = \frac{2\sqrt{1-x^2}}{\pi}, \quad -1 \leq x \leq 1.$$

### ⚠ Warning

Even though the **joint** density is perfectly flat (uniform), the **marginal** density of  $X$  alone is *not* uniform — it's largest at  $x = 0$  (where the disk is widest) and tapers to 0 at  $x = \pm 1$ . Marginalizing a uniform joint density does not generally give a uniform marginal!

### 2.1.2 Example: Buffon's needle

A floor is ruled with parallel lines a distance  $d$  apart. A needle of length  $L \leq d$  is dropped at random. What's the probability it crosses a line?

**Setup.** Let  $X$  = distance from the needle's center to the nearest line, and  $\Theta$  = acute angle the needle makes with the lines. A "random drop" means

$$X \sim \text{Uniform}(0, d/2), \quad \Theta \sim \text{Uniform}(0, \pi/2), \quad X, \Theta \text{ independent,}$$

so the joint PDF is  $f_{X,\Theta}(x, \theta) = \frac{1}{(d/2)} \cdot \frac{1}{(\pi/2)} = \frac{4}{\pi d}$  on the rectangle  $[0, d/2] \times [0, \pi/2]$ .

The needle **crosses a line** exactly when  $X \leq \frac{L}{2} \sin \Theta$  (the needle's half-length, projected perpendicular to the lines, reaches past the center to the line). So

$$P(\text{cross}) = \iint_{\{x \leq \frac{L}{2} \sin \theta\}} \frac{4}{\pi d} dx d\theta = \frac{4}{\pi d} \int_0^{\pi/2} \frac{L}{2} \sin \theta d\theta = \frac{4}{\pi d} \cdot \frac{L}{2} \cdot 1 = \frac{2L}{\pi d}.$$

### i Note

Solving for  $\pi$ :  $\pi = \frac{2L}{d \cdot P(\text{cross})}$  — dropping needles gives a (very slow!) randomized way to estimate  $\pi$ .

```
L, d = 1.0, 2.0
p_cross_formula = 2 * L / (np.pi * d)
print(f"P(cross), formula = {p_cross_formula:.5f}")

n = 2_000_000
X = rng.uniform(0, d / 2, size=n)
Theta = rng.uniform(0, np.pi / 2, size=n)
crosses = X <= (L / 2) * np.sin(Theta)
p_cross_sim = crosses.mean()
```

```
pi_estimate = 2 * L / (d * p_cross_sim)

print(f"P(cross), simulated = {p_cross_sim:.5f} (n = {n:,} drops)")
print(f"implied estimate of pi = {pi_estimate:.4f} (true pi = {np.pi:.4f})")
```

```
P(cross), formula = 0.31831
P(cross), simulated = 0.31821 (n = 2,000,000 drops)
implied estimate of pi = 3.1426 (true pi = 3.1416)
```

## 2.2 Joint CDFs

**i** Definition: Joint CDF

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u,v) dv du.$$

Same idea as a single-variable CDF: “accumulated probability,” now over a **quadrant**  $(-\infty, x] \times (-\infty, y]$  rather than a ray.

### 2.2.1 Example: uniform on the unit square

Let  $(X,Y)$  be uniform on  $[0,1]^2$ , so  $f_{X,Y}(x,y) = 1$  on the square (note: here  $X,Y$  are independent Uniform(0,1)’s, unlike the disk example). For  $0 \leq x,y \leq 1$ ,

$$F_{X,Y}(x,y) = \int_0^x \int_0^y 1 dv du = xy.$$

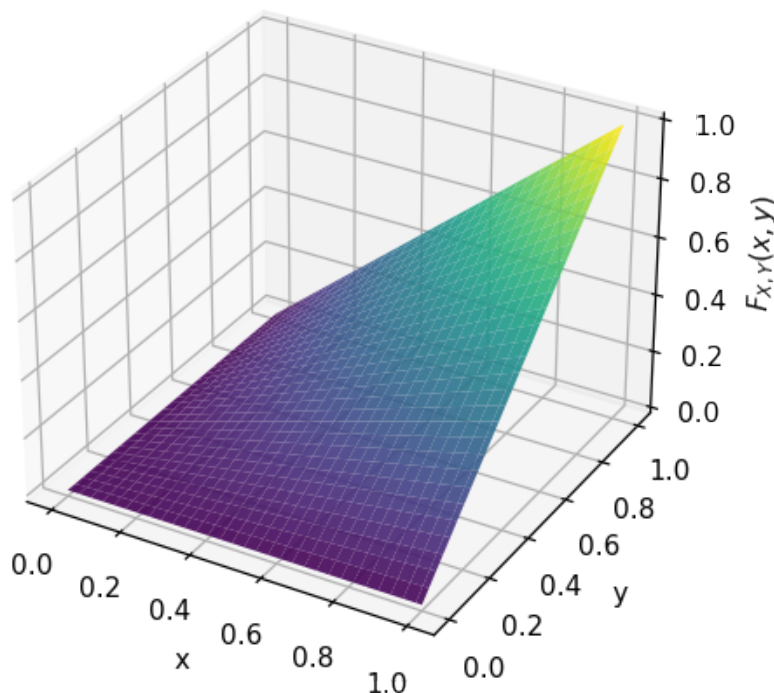
**Visual interpretation:** since the density is flat (height 1),  $F_{X,Y}(x,y)$  is just the probability mass in the rectangle  $[0,x] \times [0,y]$ , which — because density is 1 — equals the *area* of that rectangle,  $x \cdot y$ .

```
# Visualize F(x,y) = xy over the unit square, and check it against numerical integration of f=1
xg = np.linspace(0, 1, 60)
yg = np.linspace(0, 1, 60)
Xg, Yg = np.meshgrid(xg, yg)
Fg = Xg * Yg

fig = plt.figure(figsize=(6, 5))
ax = fig.add_subplot(projection="3d")
ax.plot_surface(Xg, Yg, Fg, cmap="viridis", alpha=0.9)
ax.set_xlabel("x"); ax.set_ylabel("y"); ax.set_zlabel(r"$F_{X,Y}(x,y)$")
ax.set_title("Joint CDF of Uniform on the unit square")
plt.show()

# spot check at one point via double integration of the density
x0, y0 = 0.3, 0.7
F_numeric, _ = dblquad(lambda v, u: 1.0, 0, x0, 0, y0)
print(f"F(0.3, 0.7) via dblquad = {F_numeric:.4f} (formula x*y = {x0*y0:.4f})")
```

## Joint CDF of Uniform on the unit square



$F(0.3, 0.7)$  via `dblquad` = 0.2100 (formula  $x*y = 0.2100$ )

## 2.3 Expectation for jointly continuous RVs

**i** Definition: Two-variable LOTUS

$$E[g(X, Y)] = \iint_{\mathbb{R}^2} g(x, y) f_{X,Y}(x, y) dx dy.$$

This is the continuous counterpart of the two-variable LOTUS for discrete RVs.

**💡** Fact: Linearity of expectation

For any jointly continuous  $X, Y$  (independent or not!) and constants  $a, b, c$ ,

$$E[aX + bY + c] = aE[X] + bE[Y] + c.$$

Again, this needs **no independence assumption** — it follows immediately by taking  $g(x, y) = ax + by$  in the two-variable LOTUS and using linearity of the double integral.

## 2.4 Extending to more than two random variables

Everything above extends directly to a collection  $X_1, \dots, X_n$ : a joint PDF  $f_{X_1, \dots, X_n}$  on  $\mathbb{R}^n$  that integrates to 1, marginals obtained by integrating out the variables you don't want, a joint CDF, an  $n$ -variable LOTUS, and linearity of expectation for *any* linear combination  $\sum_i a_i X_i$  — again with no independence required.

**i** A first look: multivariate Gaussian random variables

A random vector  $(X_1, \dots, X_n)$  is (jointly, or **multivariate**) **Gaussian** if every linear combination  $\sum_i a_i X_i$  is a (univariate) normal random variable. We write  $X \sim N(\mu, \Sigma)$ , where  $\mu = (E[X_1], \dots, E[X_n])$  and  $\Sigma$  is the **covariance matrix** — a natural generalization of  $\sigma^2$  that also records how the  $X_i$  vary *together*.

Two facts we'll dig into more soon:

- each individual  $X_i$  is itself normal (a marginal of a multivariate Gaussian is Gaussian);
- unlike most joint distributions, the *entire* joint distribution of a multivariate Gaussian depends on only the means and the covariance matrix  $\Sigma$ .

```
# A visual teaser: a 2D Gaussian with correlated components
rho = 0.6
mean = [0, 0]
cov = [[1, rho], [rho, 1]]

x, y = np.mgrid[-3:3:.05, -3:3:.05]
pos = np.dstack((x, y))
rv = stats.multivariate_normal(mean, cov)

plt.figure(figsize=(5.5, 5))
plt.contourf(x, y, rv.pdf(pos), levels=20, cmap="viridis")
plt.gca().set_aspect("equal")
plt.xlabel(r"$x_1$"); plt.ylabel(r"$x_2$")
plt.title(fr"2D Gaussian density, correlation $\rho$={rho}")
plt.colorbar(label="density")
plt.show()
```

